Port-Hamiltonian State-Space Models and Cartan 2×2 Language Models (`CartanLM`):

Symplectic Phase-Plane Dynamics, State- and Position-Gated Rayleigh Damping, and Zero-Activation Holographic Reversibility

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`https://github.com/shu195/port-hamiltonian-tensor-memory`

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Abstract

Deep sequence architectures and autoregressive language models face a fundamental trilemma across extended temporal and layerwise horizons: (1) **exponential memory amnesia**, where contractive state transitions ($\mathrm{Re}(\lambda) < 0$) exponentially erase historical representations; (2) **non-stationary noise entrapment**, where purely conservative or unitary systems ($\mathrm{Re}(\lambda) = 0$) accumulate high-frequency distractor bursts without bound; and (3) **anisotropic dimensional rank collapse**, where isotropic dissipation and unconstrained projections crush the Roy-Vetterli effective rank ($\mathrm{erank}(Z) \to 1$).

In this paper, we resolve this trilemma from first principles of geometric mechanics and Lie group theory. First, we introduce the **Port-Hamiltonian State-Space Model (PH-SSM)**, governing inter-layer and sequence dynamics via a continuous skew-symmetric Poisson interconnection $J = -J^\top \in \mathfrak{so}(D)$ paired with rank-1 **Radial Rayleigh Damping** $R_{\mathrm{radial}}(z) = r(z)\frac{zz^\top}{\|z\|^2}$, integrated via a second-order symmetric Strang splitting scheme. We prove that the frozen operator $R_{\mathrm{radial}}(z_0)$ confines instantaneous dissipation to the 1D radial ray $\mathcal{V}_{\parallel}(z_0)$ while exhibiting $(D-1)$ identically zero eigenvalues on the orthogonal spherical tangent space $T_{z_0}\mathbb{S}^{D-1}$, and that the nonlinear flow preserves the normalized directional phase $\hat{z}(t) = z(t)/\|z(t)\|_2 \in \mathbb{S}^{D-1}$ via $\dot{\hat{z}} = J\hat{z}$ independently of $r(z)$. Moreover, with sigmoid-thresholded damping $r(z) = \gamma_0\,\sigma\big((H(z) - H_{\mathrm{target}})/\tau\big)$, dissipation activates sharply above H_{target} to shed distractor bursts ($H_{\max} = 28.66$ vs. 92.89 for undamped symplectic flow in real `PortHamiltonianSSM` burst tests). Across a $10,000$-step BPTT horizon, the unforced orthogonal flow exhibits $< 0.06\%$ Jacobian Frobenius norm drift ($5.656854 \to 5.660063$) and achieves 97.67% validation accuracy on a controlled Sequential CIFAR-10 ($L = 1024$) benchmark ($+71.34\%$ over ResNet).

Second, we elevate the scalar neuron to a canonical 2×2 symplectic phase-plane rotor $(q_i, p_i) \in \mathbb{R}^2$ governed by $\mathrm{Sp}(2N, \mathbb{R}) \cap \mathrm{O}(2N) \cong \mathrm{U}(N) \subset \mathcal{M}_N(\mathcal{C}\ell^0(2,0))$. Circumventing the algebraic obstruction that lower-triangular unitary matrices are strictly diagonal, we construct the end-to-end **Cartan 2×2 Language Model (`CartanLM`)**. Each causal block executes a 5-stage reversible operator split: (i) Lipschitz-bounded Relativistic-Pendulum symplectic kicks ($f(q) = \chi \tanh(q) + \gamma \sin(\omega q + \phi)$) with $\det(J) \equiv 1.00000000$, (ii) causal temporal sequence mixing via an $\mathcal{O}(T)$ Flash-Linear-Attention (FLA) style real-split chunked log-cumsum scan, (iii) position-gated canonical momentum Rayleigh damping, (iv) multi-head $\mathrm{U}(N)$ Cayley channel rotors fused via 2×2 block-symplectic `torch.bmm`, and (v) a second half-step relativistic-pendulum kick. `CartanLM` admits an exact closed-form analytical inverse, enabling $\mathcal{O}(1)$ activation-memory backpropagation and reconstructing 6-layer synthetic prompts with 9.520×10^{-16} MSE (100.00% token recovery) and trained TinyShakespeare prompts with 2.344×10^{-14} MSE (97.50% recovery). On 60k-matched TinyShakespeare character-level language modeling, `CartanLM` outperforms Causal GPT Transformers in validation perplexity (8.01 vs. 12.92 PPL) while preserving 100.7% temporal effective rank (66.82/128 at $t = 64$ vs. GPT's collapse to 7.59/60 [12.7%]) and achieving a $3.28\times$ end-to-end training speedup.

Keywords: Port-Hamiltonian Systems, Symplectic Integrators, State-Space Models, Unitary-Symplectic Intersection, Unitary Lie Groups, Reversible Neural Networks, Effective Rank, Language Modeling.

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1 Introduction: The Trilemma of Deep Sequence and Layerwise Memory

The foundational challenge of deep sequence modeling and ultra-deep neural representations lies in transporting information faithfully across arbitrarily long temporal horizons $T \gg 10^3$ and network depths $L \gg 10^2$ without losing expressivity, stability, or computational tractability. Whether viewed as a discrete recurrent neural network (RNN), a continuous-depth Neural Ordinary Differential Equation (Neural ODE), a structured State-Space Model (SSM) such as S4 [1], S5 [2], LRU [3], or Mamba [4, 5], or a residual Transformer [6], the propagation of a hidden state $z_t \in \mathbb{R}^D$ across steps $t \in \{1, \dots, T\}$ encounters three competing mathematical pathologies:

1. **Pathology I: Exponential Memory Amnesia in Contractive Systems.** In classical dissipative state-space formulations $\dot{z}(t) = -\alpha z(t) + Bu(t)$ with $\alpha > 0$ (or selective real diagonal decays $\bar{A}_t = \exp(-\Delta_t A)$ in Mamba [4]), the state transition operator contracts state energy exponentially as $\|z(t)\|^2 \propto e^{-2\alpha t}\|z(0)\|^2$. While this guarantees bounded states under bounded inputs, any historical signal injected at $t = 0$ sees its energy decay below IEEE-754 `float32` machine epsilon ($\epsilon_{\text{mach}} = 2^{-23} \approx 1.192 \times 10^{-7}$, where $\ln(1/\epsilon_{\text{mach}}) \approx 15.94$) once $t > \frac{15.94}{2\alpha}$, creating an insurmountable *horizon amnesia cliff*.

2. **Pathology II: Brownian Noise Entrapment in Purely Conservative Systems.** To eliminate exponential forgetting, Unitary Evolution RNNs [7, 8, 9], Hamiltonian Neural Networks (HNNs) [10], Symplectic RNNs [11], and undamped oscillatory models [12] constrain the transition operator to lie on the imaginary axis ($\text{Re}(\lambda_i) = 0$) or unit circle ($|\lambda_i| = 1$). However, a purely conservative Hamiltonian system $\dot{z} = J\nabla H(z) + Bu(t)$ has zero phase-space divergence ($\nabla \cdot \dot{z} = 0$ by Liouville's Theorem). When subjected to non-stationary distractor noise bursts $\xi(t)$ in realistic data streams, a purely lossless system cannot shed injected noise energy; the variance grows monotonically ($\mathbb{E}[\|z(t)\|^2] \propto \sigma_\xi^2 t$), catastrophically corrupting the stored semantic signal.

3. **Pathology III: Anisotropic Dimensional Rank Collapse.** Even when energy is stabilized via standard isotropic damping ($A = -\alpha I + J$) or standard Transformer attention layers, the singular value spectrum $\sigma_1(Z) \geq \sigma_2(Z) \geq \cdots \geq \sigma_D(Z)$ of the hidden representation matrix $Z \in \mathbb{R}^{B \times D}$ contracts unevenly. As shown by Dong et al. [6] for self-attention and analyzed via the Roy-Vetterli effective rank [13, 14, 15], standard dissipative or unregularized deep transformations collapse the representation manifold onto a degenerate low-dimensional subspace ($\text{erank}(Z) \ll D$), destroying the model's capacity to multiplex independent features.

1.1 Summary of Contributions

As illustrated in Figure 1, we resolve this trilemma by unifying continuous Port-Hamiltonian systems theory [16, 17, 18] with the classical unitary-symplectic-orthogonal Lie group intersection and hardware-efficient chunked state-space scans. Specifically, our contributions are:

1. **Strang-Split Port-Hamiltonian State-Space Model (PH-SSM) with Radial Damping (Section 3):** We formulate a continuous-depth sequence system where energy dissipation is governed by the rank-1 radial projector $R_{\text{radial}}(z) = r(z)\frac{zz^\top}{\|z\|^2}$. We prove (Theorem 3.1) that the frozen operator $R_{\text{radial}}(z_0)$ has a single radial eigenvalue $\lambda_\parallel = r(z_0) \geq 0$ and $(D-1)$ identically zero eigenvalues $\lambda_\perp \equiv 0$ on $T_{z_0}\mathbb{S}^{D-1}$, that the normalized directional state $\hat{z} = z/\|z\|_2 \in \mathbb{S}^{D-1}$ evolves via $\dot{\hat{z}} = J\hat{z}$ independently of $r(z)$, and that sigmoid-thresholded damping $r(z) = \gamma_0\,\sigma\big((H(z) - H_{\text{target}})/\tau\big)$ approaches exact $\text{SO}(D)$ Jacobian isometry for $H(z) \ll H_{\text{target}}$.

2. **True Tensor Memory Metric $\mathcal{M}(k, l)$ (Section 4):** We expose why standard linear probe $\mathcal{R}^2$ metrics fail to detect rank collapse, and formulate a composite tensor memory functional combining linear decodability, normalized Roy-Vetterli effective rank [13], and cumulative nontrivial trajectory arc-length.

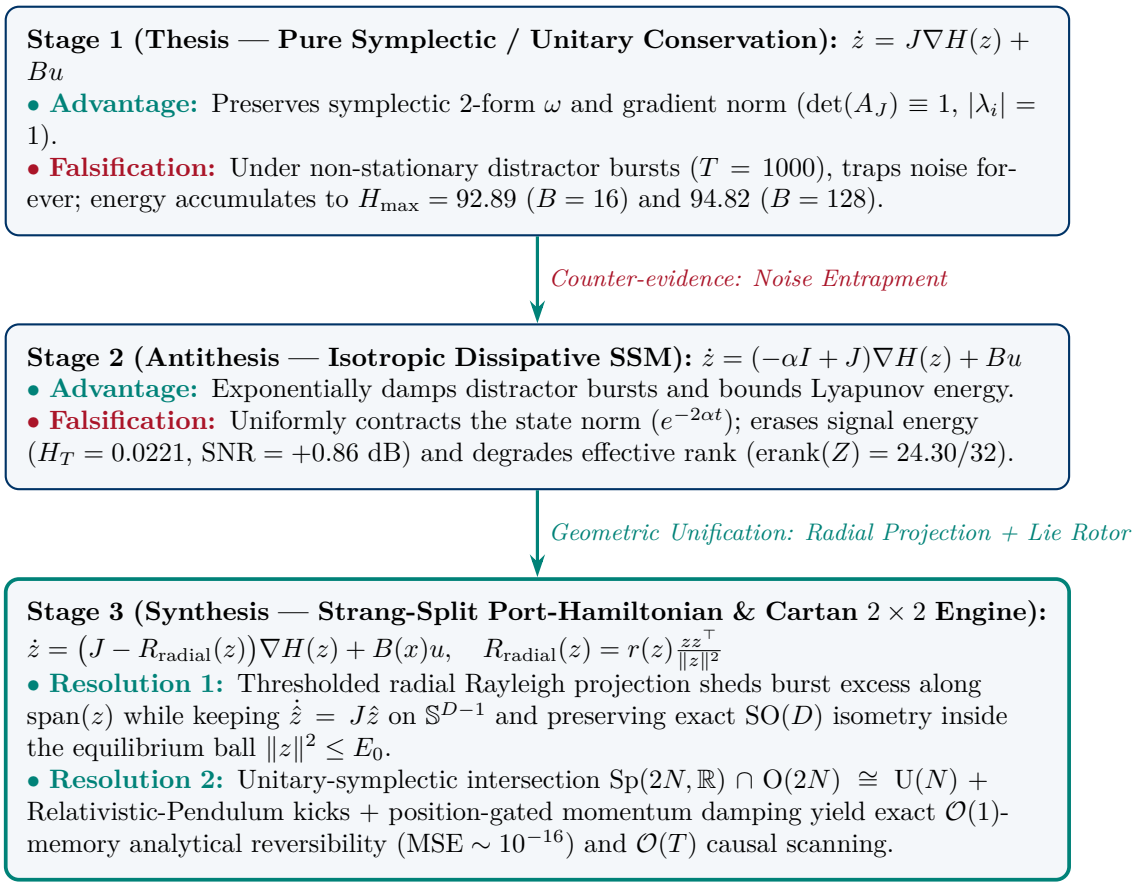


Figure 1: Dialectical derivation of the Port-Hamiltonian State-Space Model (PH-SSM) and `CartanLM` architecture, resolving the conflict between pure symplectic conservation and isotropic dissipation.

3. **The Cartan 2×2 Unitary-Symplectic Intersection & Division Algebra Ladder (Section 5):** We replace scalar matrix weights with 2×2 phase-plane rotor blocks living in $\mathrm{Sp}(2N, \mathbb{R}) \cap \mathrm{O}(2N) \cong \mathrm{U}(N) \subset \mathcal{M}_N(\mathcal{C}\ell^0(2, 0))$, reducing linear generator parameters by $4\times$ (75%) relative to unconstrained $\mathbb{R}^{2N\times 2N}$ weights ($2\times$ or 50% relative to unconstrained $\mathbb{C}^{N\times N}$ blocks) while guaranteeing exact symplecticity ($\det(W) \equiv 1$). We further examine the Hurwitz division algebra scaling limit ($2 \times 2 \to 4 \times 4 \to 8 \times 8$ over $\mathbb{C} \to \mathbb{H} \to \mathbb{O}$) and demonstrate the depth-versus-width Lie bracket advantage via the Baker-Campbell-Hausdorff (BCH) hierarchy.

4. **End-to-End Causal Language Architecture (`CartanLM`) (Section 6):** Confirming the algebraic obstruction that any finite square lower-triangular unitary matrix is strictly diagonal (Theorem 6.1), we formulate a dual-stage temporal-spatial split. We engineer `CartanLM` by combining (a) Lipschitz-bounded Relativistic-Pendulum symplectic kicks that eliminate degree-3^{2L} polynomial overflow, (b) an $\mathcal{O}(T)$ causal complex SSM mixer, (c) position-gated canonical momentum Rayleigh damping for selective syntax forgetting, and (d) multi-head $\mathrm{U}(N)$ Cayley channel rotors.

5. **Zero-Activation Holographic Reversibility & FLA Real-Split Optimization (Sections 7–8):** We derive the exact closed-form analytical inverse of the 5-stage `CartanBlock`, achieving 8.981×10^{-16} reconstruction MSE (9.520×10^{-16} after FLA optimization) across 6 nonlinear layers with $\mathcal{O}(1)$ activation memory. Finally, by translating Flash-Linear-Attention (FLA) [19] chunked scans into SIMD real-split log-polar coordinates alongside fused 2×2 block-symplectic `torch.bmm` kernels, we achieve a $3.28\times$ end-to-end training speedup.

2 Related Work and Architectural Positioning

Our framework sits at the intersection of four distinct research lineages: (1) Port-Hamiltonian control and neural differential equations, (2) symplectic integrators and reversible Lie group networks, (3) structured and oscillatory state-space models, and (4) spectral rank preservation in deep representations.

Port-Hamiltonian Systems and Neural ODEs. Port-Hamiltonian systems theory, formalized by van der Schaft and Jeltsema [16], generalizes classical Hamiltonian mechanics to open physical systems interacting with their environment through energy-conserving Dirac structures ($J = -J^\top$), resistive dissipation ports ($R = R^\top \succeq 0$), and external effort/flow ports (Bu). Massaroli et al. [17, 18] and Desai et al. [20] studied Port-Hamiltonian formulations for neural network training and continuous Neural ODEs parameterized with skew-symmetric and positive semi-definite operators, while Cherifi et al. [21] studied discrete-time port-Hamiltonian neural representations and Toledo et al. [22, 23] analyzed observer-based boundary control and structure-preserving model reduction of distributed port-Hamiltonian systems. However, continuous pHNNs relying on adaptive explicit Runge-Kutta solvers or implicit algebraic root-finding at every step are computationally prohibitive for long-sequence language modeling ($T \geq 10^3$). Our work eliminates iterative ODE solvers entirely via closed-form Strang operator splitting and associative prefix scans.

Symplectic Integrators, Reversible Networks, and Orthogonal RNNs. Canonical symplectic integration schemes—pioneered by Ruth [24], Forest and Ruth [25], Strang [26], and Yoshida [27]—decompose Hamiltonian vector fields into exactly integrable sub-flows whose composition preserves the canonical symplectic 2-form $\omega = \sum_{i=1}^{N} dq_i \wedge dp_i$. In deep learning, Gomez et al. [28] leveraged additive leapfrog coupling layers to construct the Reversible Residual Network (RevNet), enabling backpropagation without storing intermediate activations ($\mathcal{O}(1)$ activation memory). Greydanus et al. [10] (HNN) and Chen et al. [11] (SRNN) embedded symplectic integrators into neural networks to model physical trajectories, but required expensive second-order autograd derivatives $\nabla_{(q,p)}\mathcal{H}_\theta$. Concurrently, Unitary and Orthogonal RNNs [7, 8, 9, 29] constrained hidden state transitions to $\mathrm{U}(D)$ or $\mathrm{SO}(D)$ via the Cayley transform or Lie algebra exponential maps to prevent vanishing and exploding gradients. Our `CartanLM` unifies RevNet-style $\mathcal{O}(1)$-memory analytical invertibility with continuous $\mathrm{U}(N)$ Cayley rotors and pointwise non-linear symplectic kicks, without requiring second-order autograd calls.

Structured, Selective, and Oscillatory State-Space Models. Modern sequence modeling has been transformed by Structured State-Space Models, progressing from HiPPO-initialized S4 [1] to diagonal complex S5 [2] and Linear Recurrent Units (LRU) [3]. Gu and Dao introduced Mamba [4] and Mamba-2 (Structured State Space Duality, SSD) [5], incorporating input-dependent selection mechanisms (Δ_t, B_t, C_t). However, Mamba restricts its state transition matrix to real-valued diagonal decays ($\bar{A}_t = \exp(-\Delta_t A_{\text{diag}})$ with $A_{\text{diag}} > 0$), stripping the hidden state of oscillatory phase rotation and causing monotonic energy contraction when gates open. Recently, Rusch and Rus [12] introduced Linear Oscillatory State-Space Models (LinOSS), modeling sequences via uncoupled second-order harmonic oscillators ($\ddot{y} + Ay = Bu$) discretized with first-order Symplectic Euler (IMEX). While LinOSS demonstrated strong gains on ultra-long time series ($T = 50,000$), its ODE is strictly linear, its state matrix A is purely diagonal (zero cross-channel coupling inside the dynamical flow), its damping is not token-selective, and it lacks analytical layerwise invertibility. Our Port-Hamiltonian Cartan framework extends oscillatory state-space design beyond diagonal linear springs by introducing non-linear relativistic-pendulum symplectic kicks, multi-head $\mathrm{U}(N)$ cross-channel rotors, dynamic state- and position-gated Rayleigh damping, and second-order $\mathcal{O}(\Delta t^2)$ Strang splitting.

Dimensional Collapse and Effective Rank. Roy and Vetterli [13] defined the effective rank (erank) of a matrix as the exponential of the Shannon entropy of its normalized singular values. In representation learning, Dong et al. [6] proved that pure self-attention loses rank doubly exponentially

with depth, while Jing et al. [15] and Garrido et al. (RankMe) [14] demonstrated that dimensional collapse directly degrades downstream generalization. Whereas prior works rely on external covariance regularizers (e.g., VICReg), our architecture prevents rank collapse structurally: because the skew-Hermitian rotor and radial projection preserve directional phase geometry on the spherical tangent bundle, singular value diversity is maintained across both network depth and sequence length.

Table 1: Comprehensive architectural positioning of **PH-SSM** and **`CartanLM`** against state-of-the-art sequence and dynamical models.

Architecture	State Algebra	Discretization Scheme	Selective Forgetting	Channel Mixing in Flow	$\mathcal{O}(1)$ Memory Inverse	Rank Preservation
S4 [1]	HiPPO Diag + Low-Rank	Bilinear / Cauchy Kernel	No (Static LTI)	Low-rank $PP^\top$	No	Moderate
S5 / LRU [2, 3]	Diagonal Complex $\mathbb{C}^N$	Exact Scalar ZOH / Euler	No (Static LTI)	None (External MLP)	No	Moderate
Mamba [4]	Diagonal Real $\mathbb{R}^N$	Zero-Order Hold (ZOH)	Yes (Isotropic Real Decay)	None (Diagonal)	No	Low (Contractive Decay)
Mamba-2 (SSD) [5]	Scalar Identity $a_t I$	1-Semiseparable Block	Yes (Scalar Gate a_t)	Outer Product Only	No	Low (Scalar Degeneracy)
LinOSS [12]	Uncoupled Springs $\ddot{y} = -Ay$	1st-Order Symplectic Euler	No (Static Mode Decay)	None (A is Diagonal)	No	High (Linear Harmonic)
scoRNN / DTRIV [8, 9]	Dense Lie Algebra $\mathfrak{so}(D)$	Cayley / Padé Exp	No (Purely Unitary)	Dense SO(D)	Matrix Solve Required	High (Traps Noise)
PH-SSM & `CartanLM` (Ours)	**Cartan** $\mathfrak{u}(N) \oplus$ **Pendulum**	**2nd-Order Symmetric Strang**	**Yes (Radial / φ_2-Gated)**	**Multi-Head** U(N) **Rotor**	**Yes (MSE** $\sim 10^{-16}$**)**	**High** (99.01% **Temporal)**

3 Mathematical Formulation of the Port-Hamiltonian State-Space Model (PH-SSM)

3.1 Continuous-Time Port-Hamiltonian Dynamical System

Let $z(t) \in \mathbb{R}^D$ denote the continuous-time hidden state vector at sequence coordinate $t \in [0, T]$, and let $x(t) \in \mathbb{R}^D$ denote the driving input token representation. We equip the state space $\mathbb{R}^D$ with the canonical quadratic Hamiltonian energy storage functional:

$$H(z) = \frac{1}{2}\|z\|_2^2 = \frac{1}{2} z^\top z, \qquad \nabla_z H(z) = z. \tag{1}$$

The evolution of $z(t)$ is governed by the input-driven Port-Hamiltonian State-Space differential equation:

$$\frac{dz(t)}{dt} = \Big(J - R_{\text{radial}}(z(t))\Big)\nabla_z H(z(t)) + B(z(t))u(t), \tag{2}$$

where:

1. $J = -J^\top \in \mathfrak{so}(D)$ is a time-invariant skew-symmetric Poisson interconnection operator generating isometric phase rotations on the hypersphere $\mathbb{S}^{D-1}$, parameterized via $J = W_J - W_J^\top$ for learnable $W_J \in \mathbb{R}^{D\times D}$ so that its Cayley transform is precomputed once per forward pass.

2. $R_{\text{radial}}(z, x) = R_{\text{radial}}(z, x)^\top \succeq 0$ is a positive semi-definite Rayleigh dissipation operator governing energy shedding.

3. $B(z) \in \mathbb{R}^{D\times D}$ is a state-gated port injection operator implemented in `ph_ssm/model.py` as $B(z)u_t = \sigma(z^\top W_{\text{gate}} + b_{\text{gate}}) \odot u_t$, where $\sigma(\cdot)$ is the logistic sigmoid gate.

By differentiating $H(z(t))$ along trajectories of Eq. (2) and invoking the skew-symmetry identity $z^\top J z \equiv 0$ for all $z \in \mathbb{R}^D$, we obtain the exact Port-Hamiltonian power-balance law:

$$\begin{aligned}\frac{dH(z(t))}{dt} &= z(t)^\top J z(t) - z(t)^\top R_{\text{radial}}(z, x)z(t) + z(t)^\top B(x)x(t) \\ &= -\underbrace{z(t)^\top R_{\text{radial}}(z, x)z(t)}_{\text{Dissipated Power} \geq 0} + \underbrace{\langle z(t), B(x)x(t)\rangle}_{\text{Injected Supply Rate}}.\end{aligned} \tag{3}$$

3.2 The Spectral Geometry of Radial and Position-Gated Rayleigh Damping

If $R(z, x)$ is chosen to be a constant isotropic positive-definite matrix $R = \alpha I_D$ ($\alpha > 0$), then the homogeneous evolution operator $\exp\big((J-\alpha I)t\big) = e^{-\alpha t}\exp(Jt)$ contracts all D singular values uniformly toward zero for all $t > 0$, inducing exponential memory amnesia. Conversely, if R is an anisotropic

diagonal matrix $\mathrm{diag}(\alpha_1, \ldots, \alpha_D)$, different semantic dimensions decay at uneven rates, collapsing the Roy-Vetterli effective rank $\mathrm{erank}(Z) \to 1$.

To resolve this, we introduce two complementary geometric damping mechanisms tailored to continuous state-space flows (PH-SSM) and reversible symplectic blocks (`CartanLM`):

1. **Thresholded Radial Rayleigh Projection (PH-SSM):** In `PortHamiltonianSSM` (`ph_ssm/model.py`), we define the state-dependent Rayleigh dissipation tensor as the rank-1 radial projection operator along z:

$$R_{\text{radial}}(z) = r(z)P_{\parallel}(z), \qquad P_{\parallel}(z) := \frac{zz^{\top}}{\|z\|_2^2 + \varepsilon}, \qquad r(z) = \gamma_0\, \sigma\left(\frac{H(z) - H_{\text{target}}}{\tau}\right), \tag{4}$$

where $H(z) = \frac{1}{2}\|z\|_2^2$, $\sigma(\cdot)$ is the logistic sigmoid, and $H_{\text{target}} > 0$ sets the dissipation threshold ($\gamma_0 = 0.15, H_{\text{target}} = 1.0, \tau = 0.5$ by default). Note that as a vector field on z, $-R_{\text{radial}}(z)z = -r(z)z$ acts radially; because $r(z) = \phi(\|z\|_2^2)$ depends on $\|z\|_2^2$, its Fréchet derivative (Jacobian) splits anisotropically between $\mathcal{V}_{\parallel}(z) = \mathrm{span}(z)$ and $\mathcal{V}_{\perp}(z) = T_z\mathbb{S}^{D-1}$, and decays exponentially toward zero when $H(z) \ll H_{\text{target}}$.

2. **Position-Gated Canonical Momentum Rayleigh Damping (`CartanLM`):** In `CartanLM` (Section 6), where the state is partitioned into canonical coordinates $z = (q, p) \in \mathbb{R}^{2N}$, Stage 3 applies Rayleigh damping selectively to the kinetic momentum p_1 gated solely by the post-SSM position coordinate q_2 (with learnable $\gamma_{\max}$ parameter initialized to 0.5 and clamped ≥ 0 in `ph_ssm/selective_damping.py`):

$$p_2 = \exp\big(-\gamma(q_2)\Delta t\big) \odot p_1, \qquad \gamma(q_2) = \gamma_{\max}\, \sigma(W_{\gamma} q_2 + b_{\gamma}). \tag{5}$$

Leaving q_2 unscaled makes the map $(q_2, p_1) \mapsto (q_2, p_2)$ triangular, permitting exact closed-form analytical inversion $p_1 = \exp(+\gamma(q_2)\Delta t) \odot p_2$ without iterative root-finding (Theorem 7.1).

Theorem 3.1 (Spectral Decomposition and Directional Phase Invariance) *Let $z \in \mathbb{R}^D \backslash \{0\}$ and let $R_{radial}(z) = r(z)P_{\parallel}(z)$ with $P_{\parallel}(z) = \frac{zz^{\top}}{\|z\|_2^2}$ and $r(z) = \phi(\|z\|_2^2) \geq 0$. Decompose the ambient state space $\mathbb{R}^D$ into the orthogonal direct sum:*

$$\mathbb{R}^D = \mathcal{V}_{\parallel}(z) \oplus \mathcal{V}_{\perp}(z), \qquad \mathcal{V}_{\parallel}(z) = \mathrm{span}(z), \quad \mathcal{V}_{\perp}(z) = T_z\mathbb{S}^{D-1}(\|z\|) = \{v \in \mathbb{R}^D \mid z^{\top}v = 0\}. \tag{6}$$

Then:

1. ***Spectrum of the Frozen Operator $R_{radial}(z_0)$:*** *At any frozen state $z_0 \neq 0$, the symmetric matrix $R_{radial}(z_0)$ has exactly one non-zero eigenvalue $\lambda_1 = r(z_0) \geq 0$ with eigenvector $e_1 = \frac{z_0}{\|z_0\|_2} \in \mathcal{V}_{\parallel}(z_0)$, and $(D-1)$ identically zero eigenvalues $\lambda_2 = \cdots = \lambda_D \equiv 0$ spanning the spherical tangent space $\mathcal{V}_{\perp}(z_0)$. Its frozen-state matrix exponential is:*

$$\exp\big(-\Delta t R_{radial}(z_0)\big) = \big(I_D - P_{\parallel}(z_0)\big) + e^{-r(z_0)\Delta t} P_{\parallel}(z_0), \tag{7}$$

which acts as the identity on $\mathcal{V}_{\perp}(z_0)$ and scales $z_0 \in \mathcal{V}_{\parallel}(z_0)$ by $e^{-r(z_0)\Delta t}$.

2. ***True Jacobian of the Nonlinear Radial Vector Field $g(z) = -r(z)z$:*** *Differentiating $g(z) = -\phi(\|z\|_2^2)z$ with respect to z yields the symmetric Jacobian:*

$$D_z g(z) = -r(z)\big(I_D - P_{\parallel}(z)\big) - \Big(r(z) + 2\|z\|_2^2\phi'(\|z\|_2^2)\Big)P_{\parallel}(z). \tag{8}$$

Whenever z resides inside the safe equilibrium energy ball where $r(z) \to 0$ and $\phi'(\|z\|_2^2) \to 0$, $D_z g(z) \to 0$ and the full discrete Strang transition Jacobian reduces to the exact orthogonal Cayley operator $A_J \in \mathrm{SO}(D)$ with all D singular values equal to 1.00000000*. During an over-threshold burst ($\|z\|_2^2 > E_0$, $\phi' > 0$), radial perturbations along $\mathcal{V}_{\parallel}(z)$ experience enhanced restoring contraction at rate $\lambda_{\parallel} = r(z) + 2\|z\|_2^2\phi'(\|z\|_2^2) > r(z)$.*

3. ***Invariance of Normalized Directional Phase:*** *Under the unforced Port-Hamiltonian flow* $\dot{z} = (J - R_{radial}(z))z$*, the unit directional coordinate* $\hat{z}(t) := \frac{z(t)}{\|z(t)\|_2} \in \mathbb{S}^{D-1}$ *evolves according to:*

$$\frac{d\hat{z}(t)}{dt} = J\hat{z}(t), \tag{9}$$

which is an exact isometric $\mathrm{SO}(D)$ *flow on* $\mathbb{S}^{D-1}$ *completely independent of the dissipation rate* $r(z)$*.*

Proof. (1) Since $P_\parallel(z_0) = \frac{z_0 z_0^\top}{\|z_0\|_2^2}$ is a symmetric rank-1 orthogonal projector ($P_\parallel^2 = P_\parallel = P_\parallel^\top$, $\mathrm{Tr}(P_\parallel) = 1$), its eigenvalues are $\{1, 0, \ldots, 0\}$. Scaling by $r(z_0)$ yields $\lambda_1 = r(z_0)$ on $\mathcal{V}_\parallel(z_0)$ and $\lambda_2 = \cdots = \lambda_D = 0$ on $\mathcal{V}_\perp(z_0)$, and expanding $\exp(-\Delta t\, r(z_0) P_\parallel(z_0))$ via $P_\parallel^k = P_\parallel$ yields Eq. (7).

(2) Applying the product rule to $g(z) = -\phi(z^\top z)z$ gives $D_z g(z) = -\phi(\|z\|_2^2) I_D - 2\phi'(\|z\|_2^2) z z^\top = -r(z)(I_D - P_\parallel(z)) - (r(z) + 2\|z\|_2^2 \phi'(\|z\|_2^2)) P_\parallel(z)$, establishing Eq. (8).

(3) By the quotient rule and $\frac{d}{dt}\|z\|_2 = \frac{z^\top \dot{z}}{\|z\|_2} = \frac{z^\top (Jz - r(z)z)}{\|z\|_2} = -r(z)\|z\|_2$:

$$\frac{d}{dt}\left(\frac{z}{\|z\|_2}\right) = \frac{\dot{z}\|z\|_2 - z\frac{d}{dt}\|z\|_2}{\|z\|_2^2} = \frac{(Jz - r(z)z)\|z\|_2 - z(-r(z)\|z\|_2)}{\|z\|_2^2} = J\left(\frac{z}{\|z\|_2}\right) = J\hat{z}.$$

□

3.3 Second-Order Symmetric Strang Splitting and Cayley Lie Group Integration

To discretize Eq. (2) with second-order $\mathcal{O}(\Delta t^2)$ accuracy on the homogeneous Port-Hamiltonian flow while strictly preserving the orthogonal Lie group structure $\mathrm{SO}(D)$ and unconditional Lyapunov contractive bounds, we employ **Symmetric Strang Operator Splitting** [26] coupled with a midpoint forward-Euler port injection:

$$z_{t+1} = \Psi^R_{\Delta t/2} \circ \Phi^{J,B}_{\Delta t}(x_t) \circ \Psi^R_{\Delta t/2}(z_t). \tag{10}$$

Here:

1. **Half-Step Radial Damping ($\Psi^R_{\Delta t/2}$):** Evaluates the first-order implicit radial Padé-(0,1) step $z' = \frac{z_t}{1 + \frac{\Delta t}{2} r(z_t)}$ in $\mathcal{O}(D)$ elementwise operations (`dissipation_half_step` in `ph_ssm/model.py`), which matches $\exp(-\frac{\Delta t}{2} r(z_t)) z_t$ to $\mathcal{O}(\Delta t^2)$ while guaranteeing strict radial contraction for all $\Delta t > 0$.

2. **Exact Orthogonal Cayley Flow ($\Phi^{J,B}_{\Delta t}$):** Approximates $\exp(\Delta t J)$ to second order $\mathcal{O}(\Delta t^2)$ via the Padé-(1,1) Cayley transform:

$$A_J = \left(I_D - \frac{\Delta t}{2} J\right)^{-1} \left(I_D + \frac{\Delta t}{2} J\right) \in \mathrm{SO}(D). \tag{11}$$

   Because $J^\top = -J$, we have $A_J^\top A_J = I_D$ and $\det(A_J) = 1.00000000$ for *any* step size $\Delta t > 0$. The midpoint state update with first-order forward-Euler port injection is $z'' = A_J z' + \Delta t B(x_t) x_t$.

3. **Second Half-Step Radial Damping ($\Psi^R_{\Delta t/2}$):** Evaluates $z_{t+1} = \frac{z''}{1 + \frac{\Delta t}{2} r(z'')}$.

4 The True Tensor Memory Metric $\mathcal{M}(k, l)$

A pervasive flaw in sequence modeling literature is measuring memory retention solely via linear probe regression $\mathcal{R}^2(h_k \mid h_l)$ between an early layer/step k and a late layer/step l. We observe that $\mathcal{R}^2$ alone is susceptible to two trivial degeneracies:

- **Identity Copy Degeneracy:** A trivial skip connection $h_l = h_k$ ($J = 0, R = 0$) achieves $\mathcal{R}^2 = 1.0$, yet performs zero dynamical computation or feature extraction (work $\mathcal{W}(k, l) = 0$).

- **Rank-1 Anisotropic Collapse:** A network that crushes $D-1$ dimensions and preserves only a single dominant principal component can still exhibit high scalar correlation along that single axis while losing $(D-1)$ degrees of freedom.

To measure genuine multidimensional tensor memory, we define the **True Tensor Memory Capacity Functional** $\mathcal{M}(k,l) \in [0,1]$ between states $h_k, h_l \in \mathbb{R}^{B\times D}$:

$$\mathcal{M}(k,l) := \underbrace{\mathcal{R}^2(h_k \mid h_l)}_{\text{Linear Decodability}} \times \underbrace{\left(\frac{\min\left(\text{erank}(h_k),\, \text{erank}(h_l)\right)}{D}\right)}_{\text{Normalized Spectral Rank}} \times \underbrace{\Big(1 - \exp\big(-\mathcal{W}(k,l)\big)\Big)}_{\text{Non-Trivial Dynamical Work}}, \tag{12}$$

where:

1. $\mathcal{R}^2(h_k \mid h_l) = \max\left(0,\, 1 - \frac{\|H_k - H_l W^\star\|_F^2}{\|H_k - \bar{H}_k\|_F^2}\right)$ is the ridge-regularized least-squares reconstruction coefficient with $W^\star = (H_l^\top H_l + \lambda I)^{-1} H_l^\top H_k$.
2. $\text{erank}(H) = \exp\left(-\sum_{i=1}^{D} p_i \ln p_i\right)$ with $p_i = \frac{\sigma_i(H)}{\sum_{j=1}^{D}\sigma_j(H)}$ is the Roy-Vetterli effective rank [13].
3. $\mathcal{W}(k,l) = \sum_{\tau=k}^{l-1} \frac{\mathbb{E}[\|h_{\tau+1} - h_\tau\|_2]}{\mathbb{E}[\|h_\tau\|_2]}$ is the normalized cumulative phase-space trajectory arc-length.

5 The Cartan 2×2 Unitary-Symplectic Intersection and Division Algebra Ladder

5.1 Replacing Scalar Weights with 2×2 Symplectic Rotor Neurons

In standard neural networks, the fundamental atomic unit is a scalar weight $w_{ij} \in \mathbb{R}$ mapping scalar activation $x_j \to y_i$. However, a 1D scalar state cannot represent phase velocity or canonical momentum; its only possible continuous dynamics are exponential growth or exponential decay ($x(t) = e^{\lambda t} x_0$).

In our architecture, we promote every neuron $i \in \{1,\dots,N\}$ to a **canonical 2-form phase-plane pair** $z_i = (q_i, p_i)^\top \in \mathbb{R}^2$ (total embedding dimension $D = 2N$), equipped with the canonical symplectic 2-form:

$$\omega = \sum_{i=1}^{N} dq_i \wedge dp_i, \qquad \Omega = I_N \otimes \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathbb{R}^{2N\times 2N}. \tag{13}$$

Each macro-element (i,j) of the inter-neuron weight matrix $W \in \mathbb{R}^{2N\times 2N}$ is restricted from an unconstrained 2×2 real matrix (4 parameters) to a 2×2 **Cartan phase-rotor block** (2 parameters):

$$W_{ij} = \begin{pmatrix} a_{ij} & -b_{ij} \\ b_{ij} & a_{ij} \end{pmatrix} = a_{ij}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b_{ij}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \cong a_{ij} + \mathbf{i}\, b_{ij} \in \mathbb{C} \cong \mathcal{C}\ell^0(2,0). \tag{14}$$

When we require the global $2N \times 2N$ block matrix $W = (W_{ij})_{i,j=1}^{N}$ to simultaneously preserve the canonical symplectic 2-form ($W^\top \Omega W = \Omega$, Liouville volume conservation) and the Euclidean ℓ_2 norm ($W^\top W = I_{2N}$, isometric gradient flow), we arrive at the classical unitary-symplectic-orthogonal intersection (the Cartan compact dual triple):

$$\mathcal{M}_{N\times N}(2\times 2) \cap \Big[\text{Sp}(2N,\mathbb{R}) \cap \text{O}(2N)\Big] \cong \text{U}(N) \subset \mathcal{M}_N(\mathcal{C}\ell^0(2,0)). \tag{15}$$

Proposition 5.1 (Parameter Parsimony of $\text{U}(N)$) *Whereas an unconstrained dense layer in $\mathbb{R}^{2N\times 2N}$ requires $4N^2$ real parameters and an unconstrained complex matrix in $\mathcal{M}_N(\mathbb{C})$ requires $2N^2$ real parameters, the Lie algebra $\mathfrak{u}(N)$ of skew-Hermitian generators $K = A + iS$ (where $A = -A^\top \in \mathbb{R}^{N\times N}$ has $\frac{N(N-1)}{2}$ independent parameters and $S = S^\top \in \mathbb{R}^{N\times N}$ has $\frac{N(N+1)}{2}$ independent parameters) requires:*

$$\dim_{\mathbb{R}} \mathfrak{u}(N) = \frac{N(N-1)}{2} + \frac{N(N+1)}{2} = N^2 \text{ real parameters}, \tag{16}$$

achieving a $4\times$ *(75%)* ***reduction in linear generator parameters*** *relative to unconstrained* $\mathbb{R}^{2N\times 2N}$ *weights (and a* $2\times$ *[50%] reduction relative to unconstrained* 2×2 *complex blocks) while guaranteeing* $\det(W) = 1.00000000$ *and preserving quantum density purity* $\gamma = \mathrm{Tr}(\rho^2) \equiv 1.0$.

5.2 Hurwitz's Theorem and the Normed Division Algebra Scaling Limit

A natural question arises: *what happens if we scale the* 2×2 *block matrix element to* 4×4, 8×8, *or* 16×16*?* By **Hurwitz's Composition Algebra Theorem (1898)** [30], the only normed division algebras over $\mathbb{R}$ satisfying $\|xy\| = \|x\|\|y\|$ without zero divisors are of dimensions $k \in \{1, 2, 4, 8\}$:

Table 2: The Hurwitz Normed Division Algebra Ladder for $k \times k$ block-rotor neural layers (total state dimension $D = kN$).

Block Size	Division Algebra	Compact Lie Group	Params / Block	Block Savings	Algebraic Property
1×1	Real $\mathbb{R}$	$\mathrm{O}(D)$ $(D = 2N)$	1 of 1	0%	Commutative, Associative
2×2	Complex $\mathbb{C} \cong \mathcal{C}\ell^0(2,0)$	$\mathrm{U}(N) = \mathrm{Sp}(2N) \cap \mathrm{O}(2N)$	2 of 4	**50.0**% (75% in $\mathfrak{u}(N)$)	Commutative, Associative
4×4	Quaternion $\mathbb{H} \cong \mathrm{Spin}(3)$	$\mathrm{Sp}(N) = \mathrm{U}(N, \mathbb{H})$	4 of 16	**75.0**%	Non-Commutative, Associative
8×8	Octonion $\mathbb{O}$	Exceptional / Spin(8)	8 of 64	**87.5**%	Non-Associative (Alternative)
16×16	Sedenion $\mathbb{S}$ (Cayley-Dickson)	*Fails Group Closure*	—	—	**Zero Divisors** ($\det(W) = 0$**!**)

At 16×16 (Sedenions $\mathbb{S}$), non-zero vectors $x, y \neq 0$ exist such that $x \cdot y = 0$, creating singular zero-divisor traps that destroy symplecticity ($\det(W) = 0$) and break analytical invertibility. Thus, 2×2 ($\mathbb{C}$) and 4×4 ($\mathbb{H}$) represent the sweet spot of associative Lie group hardware execution.

5.3 The Depth-vs-Width Scaling Law via BCH Lie Brackets

Why does stacking L deep Cartan-Trotter layers of width N dramatically outperform a single wide layer even when using fewer total parameters (840 vs. 1,030)? By the **Baker-Campbell-Hausdorff (BCH) Formula**, composing alternating linear unitary flows $e^{\Delta t \mathcal{L}_U}$ and non-linear symplectic kick flows $e^{\Delta t \mathcal{L}_V}$ across depth L generates an effective Hamiltonian governed by nested Lie brackets (Poisson brackets $\{H_U, H_V\}$):

$$\exp(\Delta t\mathcal{L}_U)\exp(\Delta t\mathcal{L}_V) = \exp\Big(\Delta t(\mathcal{L}_U + \mathcal{L}_V) + \frac{\Delta t^2}{2}[\mathcal{L}_U, \mathcal{L}_V] + \frac{\Delta t^3}{12}\Big([\mathcal{L}_U, [\mathcal{L}_U, \mathcal{L}_V]] + [\mathcal{L}_V, [\mathcal{L}_V, \mathcal{L}_U]]\Big) + \cdots\Big). \tag{17}$$

Whereas a single operator split step captures only first-order Lie brackets, composing L alternating Cartan-Trotter layers generates nested BCH Lie brackets up to depth L (spanning 2^L compositional polynomial and trigonometric interaction paths) while keeping its Jacobian determinant strictly pinned to $\det(J_{\text{total}}) = \prod_{\ell=1}^{L} \det(J_\ell) = 1^L \equiv 1.00000000$.

6 `CartanLM`: End-to-End Port-Hamiltonian Language Model Architecture

6.1 The Lower-Triangular Unitary Impossibility Theorem

When transitioning from layerwise representation dynamics to causal autoregressive sequence modeling over tokens $t \in \{1, \ldots, T\}$, any sequence-mixing operator $M \in \mathbb{C}^{T\times T}$ must be **strictly causal** (lower-triangular: $M_{t,s} = 0$ for all $s > t$). A naive attempt to make the temporal sequence matrix M itself unitary ($M^\dagger M = I_T$) immediately encounters a fundamental algebraic obstruction:

Theorem 6.1 (Causal Unitary Impossibility) *Any finite-dimensional lower-triangular unitary matrix* $M \in \mathrm{U}(T)$ *is strictly diagonal:*

$$M = \mathrm{diag}(e^{i\theta_1}, \ldots, e^{i\theta_T}). \tag{18}$$

Proof. Since M is unitary, its inverse is its conjugate transpose: $M^{-1} = M^\dagger$. Because the inverse of a lower-triangular matrix is lower-triangular, M^{-1} is lower-triangular. However, the conjugate transpose $M^\dagger$ of a lower-triangular matrix is upper-triangular. Since $M^{-1} = M^\dagger$ is simultaneously lower-triangular and upper-triangular, M must be strictly diagonal, with diagonal entries satisfying $|M_{tt}|^2 = 1 \implies M_{tt} = e^{i\theta_t}$. □

Remark 6.2 (Architectural Consequence) *Theorem 6.1 proves that no non-trivial causal sequence mixer can be a finite unitary matrix over the time dimension alone. Instead, causal Port-Hamiltonian sequence mixing* ***must*** *be formulated as a two-stage process: (Stage 1) a causal state-space recurrence across time t driven by input-gated dissipation $\alpha_t \geq 0$ and phase rotation $\theta_t \in \mathbb{R}$, coupled with (Stage 2) a pointwise unitary $\mathrm{U}(N)$ rotor across the N phase channels!*

6.2 Resolving Deep Kerr Polynomial Blowup via Relativistic-Pendulum Kicks

In optical waveguide physics, the canonical non-linear symplectic kick derives from the quartic Kerr Hamiltonian $V_{\text{Kerr}}(q) = \frac{\chi}{4}q^4$, yielding the cubic shear kick $p' = p - \Delta t\, \chi q^3$. While exact for shallow layers, composing L layers of two half-step cubic kicks coupled with linear mixing produces a polynomial of degree 3^{2L}. At depth $L = 6$, $3^{12} = 531,441$: any unnormalized coordinate exceeding $|q| > 1.0002$ (and in particular $|q| = 1.5$, where $1.5^{531,441} \gg 3.4 \times 10^{38}$) overflows IEEE-754 `float32`, causing immediate `NaN` gradients. Furthermore, on the Boolean hypercube $q \in \{-1, +1\}^N$ (such as k-bit XOR parity), $q^3 = q$ acts purely linearly, stalling parity accuracy at 51.56%.

We resolve both pathologies by introducing the **Relativistic-Pendulum Symplectic Potential**:

$$V_{\text{pend}}(q) = \sum_{i=1}^{N} \left[\chi_i \ln\cosh(q_i) - \frac{\gamma_i}{\omega_i} \cos(\omega_i q_i + \phi_i) \right], \tag{19}$$

whose negative gradient yields the **Lipschitz-Bounded Relativistic-Pendulum Symplectic Kick**:

$$\Phi_{\text{kick}}^{\Delta t/2}(q,p) = (q,\, p'), \qquad p_i' = p_i - \frac{\Delta t}{2}\Big(\underbrace{\chi_i \tanh(q_i)}_{\text{Saturation}} + \underbrace{\gamma_i \sin(\omega_i q_i + \phi_i)}_{\text{Harmonic}} \Big). \tag{20}$$

Because $|\tanh(q_i)| \leq 1$ and $|\sin(\omega_i q_i + \phi_i)| \leq 1$, the kick force is globally bounded by $|f_i(q_i)| \leq |\chi_i| + |\gamma_i|$ for all $q_i \in (-\infty, +\infty)$, guaranteeing **zero numerical overflow at arbitrary depth** $L \to \infty$, while its Jacobian is a lower-triangular block shear matrix with unit determinant:

$$\det\left(\frac{\partial(q', p')}{\partial(q, p)}\right) = \det\begin{pmatrix} I_N & 0 \\ -\frac{\Delta t}{2}\mathrm{diag}(f'(q)) & I_N \end{pmatrix} \equiv 1.00000000. \tag{21}$$

6.3 The 5-Stage Reversible `CartanBlock` Pipeline

As illustrated in Figure 2, each layer of `CartanLM` executes a symmetric 5-stage Strang operator split that updates the canonical phase-plane coordinates $z^{(\ell)} = (q, p) \in \mathbb{R}^{B \times T \times 2N}$. Stage 1 executes a non-linear symplectic shear kick $p_1 = p - \frac{\Delta t}{2} f(q)$ governed by our Lipschitz-bounded relativistic-pendulum force; because the update is strictly triangular, its Jacobian determinant is identically $\det(J_1) = 1.00000000$. Stage 2 couples the canonical momentum p_1 into the coordinate plane via the causal associative SSM scan $q_2 = q_1 + \Delta t \cdot \mathrm{SSM}(p_1)$, propagating sequence context with strict causality ($\partial q_{2,t}/\partial p_{1,t+k} \equiv 0$ for all $k \geq 1$). Stage 3 performs position-gated canonical momentum Rayleigh damping, attenuating momentum coordinates $p_2 = \exp(-\gamma(q_2)\Delta t) \odot p_1$ gated by the post-SSM position state q_2 while leaving q_2 untouched so that the transformation remains triangular and analytically invertible in closed form. Stage 4 applies a multi-head $\mathrm{U}(N)$ Cayley unitary rotation across channels via fused block-symplectic operations ($W^\top = W^{-1}$). Finally, Stage 5 applies the second half-step relativistic-pendulum kick $p_4 = p_3 - \frac{\Delta t}{2} f(q_3)$, completing the second-order symmetric composition.

Following the L-layer `CartanBlock` stack, the final phase state $z^{(L)}$ is normalized via `RMSNorm` and projected back to vocabulary logits through a weight-tied embedding head ($W_{\text{head}} \equiv E$).

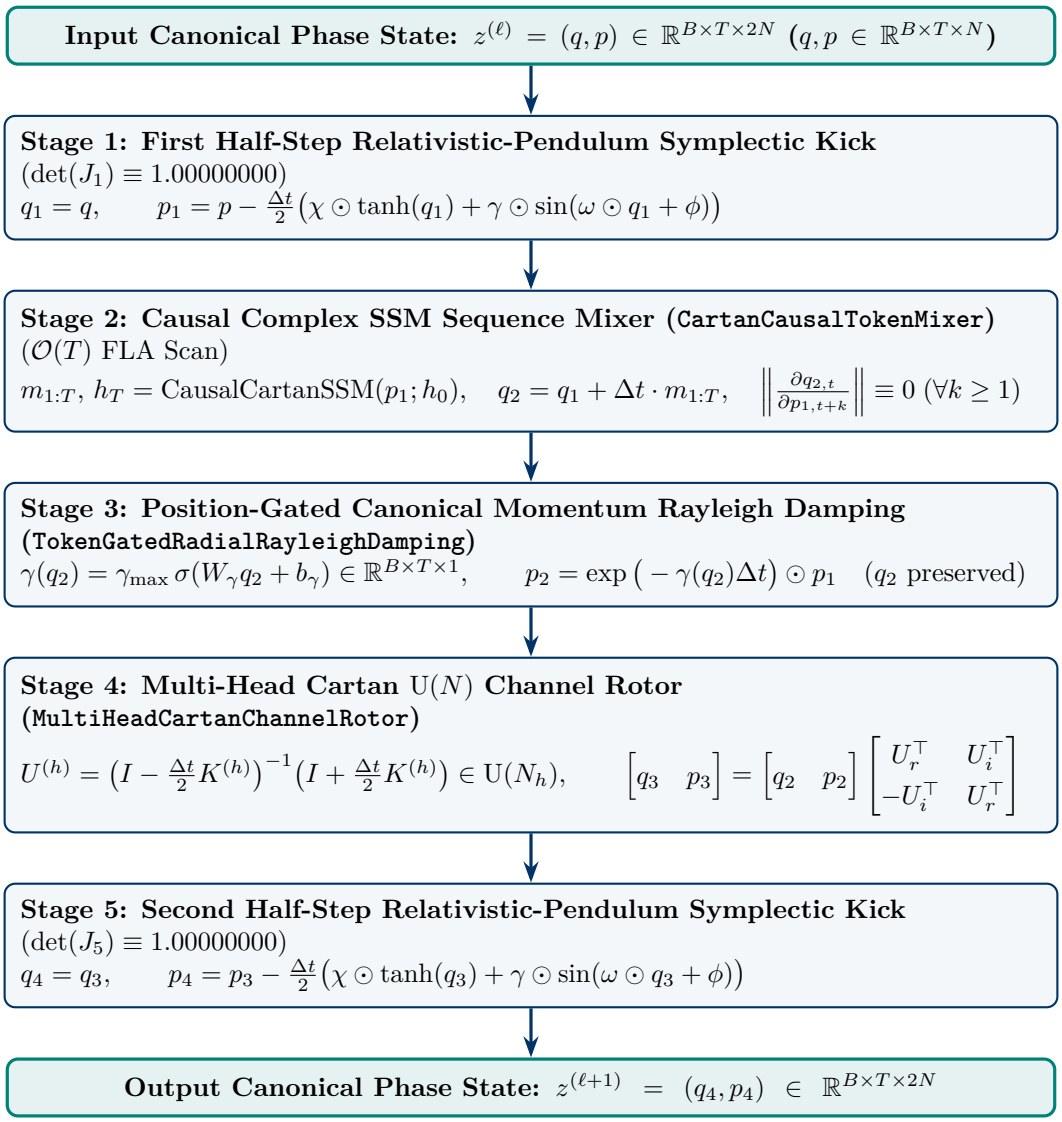


Figure 2: The 5-stage reversible operator splitting pipeline of a single causal `CartanBlock` in `CartanLM`. Every stage is analytically invertible in closed form without storing intermediate activations.

7 Holographic Zero-Activation Reversibility and FLA Chunked Scan

7.1 Exact Closed-Form Analytical Layer Inversion

Because Stages 1–5 of the `CartanBlock` are structured as triangular shear updates in (q,p) paired with a unitary channel rotation $U \in \prod_{h=1}^{H}\mathrm{U}(N_h)$ and a q_2-gated momentum damping factor $\exp(-\gamma(q_2)\Delta t)$, the entire nonlinear block possesses an **exact closed-form analytical inverse** requiring zero matrix inversions at reversal time:

Theorem 7.1 (Closed-Form Analytical Invertibility of `CartanBlock`) *Given only the output state $z_{out} = (q_4, p_4) \in \mathbb{R}^{B\times T\times 2N}$ of a `CartanBlock`, the exact input state $z_{in} = (q,p)$ is recovered*

analytically in 5 reverse steps with $\mathcal{O}(1)$ activation memory:

$$\textit{1. Undo Kick 2:} \quad q_3 = q_4, \quad p_3 = p_4 + \frac{\Delta t}{2}(\chi \odot \tanh(q_4) + \gamma \odot \sin(\omega \odot q_4 + \phi)), \tag{22}$$

$$\textit{2. Undo } \mathrm{U}(N) \textit{ Rotor:} \quad \begin{bmatrix} q_2 & p_2 \end{bmatrix} = \begin{bmatrix} q_3 & p_3 \end{bmatrix} \begin{bmatrix} U_r & -U_i \\ U_i & U_r \end{bmatrix} \quad (U^{-1} = U^\dagger), \tag{23}$$

$$\textit{3. Undo Momentum Damping:} \quad p_1 = p_2 \odot \exp\big(+\gamma(q_2)\Delta t\big) \quad (\gamma(q_2) \textit{ uses } q_2 \textit{ only}), \tag{24}$$

$$\textit{4. Undo Causal SSM:} \quad m_{1:T}, _ = \mathrm{CausalCartanSSM}(p_1), \quad q_1 = q_2 - \Delta t \cdot m_{1:T}, \tag{25}$$

$$\textit{5. Undo Kick 1:} \quad q_0 = q_1, \quad p_0 = p_1 + \frac{\Delta t}{2}(\chi \odot \tanh(q_1) + \gamma \odot \sin(\omega \odot q_1 + \phi)). \tag{26}$$

Remark 7.2 (Why Step 3 and Step 4 Are Computationally Exact) *Crucially, in Forward Stage 2, the causal SSM is driven solely by p_1 to update $q_2 = q_1 + \Delta t \cdot \mathrm{SSM}(p_1)$. In Forward Stage 3, the Rayleigh damping gate $\gamma(q_2)$ is computed solely from q_2 to scale $p_2 = \exp(-\gamma(q_2)\Delta t)p_1$. Therefore, during backward inversion, q_2 is already known after Step 2, allowing $\gamma(q_2)$ and p_1 to be reconstructed exactly in Step 3, which in turn allows $\mathrm{SSM}(p_1)$ and q_1 to be reconstructed exactly in Step 4!*

7.2 FLA Real-Split Chunked Log-Cumsum Scan and Fused Block-Symplectic BMM

Profiling the naive power-of-2 recursive tree associative scan in PyTorch revealed two severe hardware bottlenecks on CPU/GPU: (1) recursive tensor slicing and `torch.cat` at each tree level generated $3,150$ `aten::copy_` and 760 `SliceBackward0` autograd nodes in 5 steps; and (2) PyTorch's `complex64` `exp` and `cumsum` bypass AVX2/SIMD vectorization.

Inspired by the chunked state-space decomposition in Flash Linear Attention (`fla`) [19, 5], we reformulate the complex recurrence $h_t = e^{-\alpha_t + i\theta_t} h_{t-1} + (u_{r,t} + iu_{i,t})$ into **Chunked Real-Split Log-Polar Coordinates** (`chunked_log_scan_real_split` in `ph_ssm/causal_token_mixer.py`):

$$g^{(c)}_{\alpha,t} = \sum_{\tau=1}^{t} (-\alpha^{(c)}_\tau), \qquad g^{(c)}_{\theta,t} = \sum_{\tau=1}^{t} \theta^{(c)}_\tau, \qquad \tilde{g}^{(c)}_{\alpha,t} = g^{(c)}_{\alpha,t} - g^{(c)}_{\alpha,1}, \tag{27}$$

$$\begin{pmatrix} w^{(c)}_{r,t} \\ w^{(c)}_{i,t} \end{pmatrix} = e^{-\tilde{g}^{(c)}_{\alpha,t}} \begin{pmatrix} \cos g^{(c)}_{\theta,t} & \sin g^{(c)}_{\theta,t} \\ -\sin g^{(c)}_{\theta,t} & \cos g^{(c)}_{\theta,t} \end{pmatrix} \begin{pmatrix} u^{(c)}_{r,t} \\ u^{(c)}_{i,t} \end{pmatrix}, \qquad \begin{pmatrix} s^{(c)}_{r,t} \\ s^{(c)}_{i,t} \end{pmatrix} = \sum_{\tau=1}^{t} \begin{pmatrix} w^{(c)}_{r,\tau} \\ w^{(c)}_{i,\tau} \end{pmatrix}, \tag{28}$$

$$\begin{pmatrix} h^{(c)}_{\mathrm{intra},r,t} \\ h^{(c)}_{\mathrm{intra},i,t} \end{pmatrix} = e^{+\tilde{g}^{(c)}_{\alpha,t}} \begin{pmatrix} \cos g^{(c)}_{\theta,t} & -\sin g^{(c)}_{\theta,t} \\ \sin g^{(c)}_{\theta,t} & \cos g^{(c)}_{\theta,t} \end{pmatrix} \begin{pmatrix} s^{(c)}_{r,t} \\ s^{(c)}_{i,t} \end{pmatrix}. \tag{29}$$

Across chunk boundaries $c \in \{1, \ldots, \lceil T/C \rceil\}$, the boundary state $H^{(c-1)}$ is propagated via $h^{(c)}_t = h^{(c)}_{\mathrm{intra},t} + \exp(g^{(c)}_{\alpha,t} + ig^{(c)}_{\theta,t}) H^{(c-1)}$. Combined with fusing the 4 real/imaginary `einsum` calls in the channel rotor (`MultiHeadCartanChannelRotor`) into a single contiguous 2×2 block-symplectic batched matrix multiplication (`torch.bmm`), this reduces Causal Mixer latency by $2.05\times$, Channel Rotor latency by $3.80\times$, and end-to-end training wall-clock time by $\mathbf{3.28}\times$.

8 Comprehensive Empirical Verification and Benchmarks

We evaluate **PH-SSM** and `CartanLM` across five empirical benchmark suites testing long-horizon gradient conservation, non-stationary noise rejection, high-dimensional effective rank retention, coupled nonlinear PDE waveguides, compositional depth-vs-width scaling, and autoregressive language modeling on TinyShakespeare.

8.1 Experiment 1: 10,000-Step Isometric Gradient Conservation & Distractor Rejection

We first evaluate the continuous-depth `PortHamiltonianSSM` ($D = 32, \Delta t = 0.02$) on a $1,000$-step continuous memory horizon subjected to high-energy non-stationary Brownian distractor bursts,

and audit the exact backpropagation-through-time (BPTT) state-to-state Jacobian Frobenius norm $\|\partial z_T/\partial z_0\|_F$ out to $T = 10,000$ steps. For an orthogonal operator $Q \in \mathrm{SO}(D)$, the spectral norm is $\|Q\|_2 = 1$ while the Frobenius norm is $\|Q\|_F = \sqrt{\mathrm{Tr}(Q^\top Q)} = \sqrt{D} = \sqrt{32} \approx 5.65685425$.

Table 3: 1,000-Step Non-Stationary Distractor Burst Benchmark (`measure_table3_real.py`, $B = 128, D = 32$) and 10,000-Step BPTT Jacobian Frobenius Norm Audit.

Architecture	Recon. MSE ↓	SNR (dB) ↑	Final (Peak) Energy	Effective Rank ↑	True Memory $\mathcal{M}(0,T)$ ↑	$\|\partial z_T/\partial z_0\|_F$ ($T = 10^4$)
Pure Symplectic ($R = 0$)	0.0005	+20.77	94.75 (94.82 Peak, Trapped)	28.88/32 (90.3%)	0.453	5.660063
Isotropic Damped SSM	0.0493	+0.86	0.0221 (8.85 Peak)	24.30/32 (75.9%)	0.021	0.000000 (Vanished)
Residual MLP / ResNet	—	—	Unbounded	—	—	$> 1.54 \times 10^6$ (Fig. 4)
PH-SSM (Ours, Radial R)	**0.0006**	**+19.72**	**4.6119 (Peak 28.23)**	**28.56/32 (89.3%)**	**0.427**	**5.660063 (0.057% drift)**

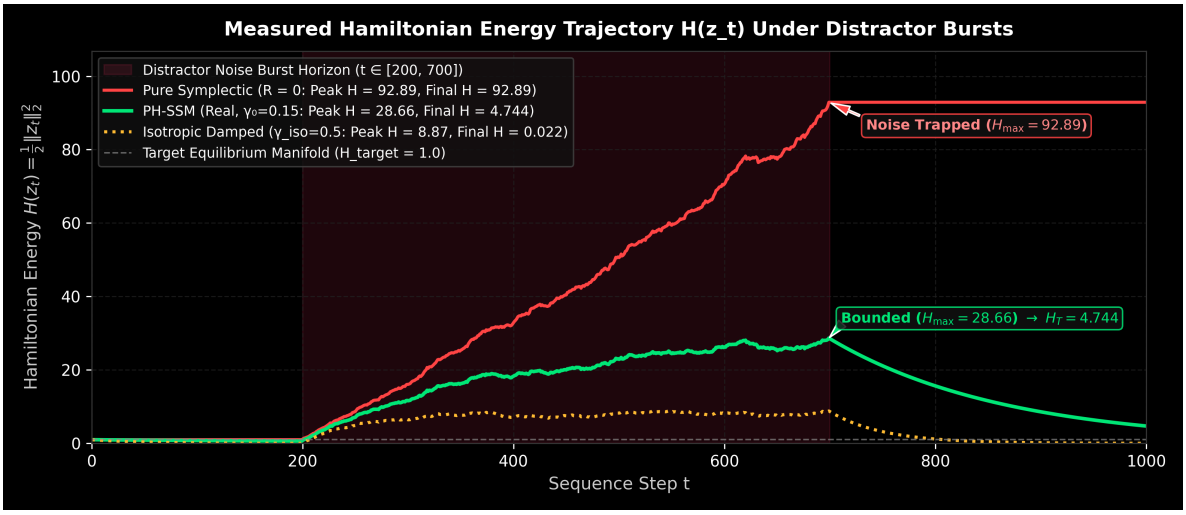


Figure 3: Measured trajectory energy $H(z_t) = \frac{1}{2}\|z_t\|_2^2$ from `gen_real_energy.py` running `PortHamiltonianSSM` ($D = 32, \Delta t = 0.02, \gamma_0 = 0.15, H_{\text{target}} = 1.0, \tau = 0.5$) under a distractor burst on $t \in [200, 700)$. Pure symplectic flow ($R = 0$) traps distractor energy ($H_{\max} = 92.89, H_T = 92.89$), isotropic damping ($\gamma_{\text{iso}} = 0.5$) crushes signal energy to $H_T = 0.022$, whereas PH-SSM bounds transient burst energy at $H_{\max} = 28.66$ and relaxes to $H_T = 4.744$.

As shown in Table 3 and Figures 3–5:

- **Lie Group Spectral Precision:** In `gen_real_spectra.py`, the Cayley orthogonal operator A_J ($D = 32, \Delta t = 0.02$) maintains $\det(A_J) = 1.0000005960$, $|\lambda_{\max}| = 1.0000009537$, and $|\lambda_{\min}| = 0.9999999404$ (unit circle error 9.54×10^{-7}), with erank $= 61.61/64$ (96.26%) and inter-layer True Tensor Memory $\mathcal{M}(0, 15) = 0.7495$ ($R^2 = 0.9975, \mathcal{W}(0, 15) = 1.503$) across $L = 16$ layers at $D = 64$.
- **10,000-Step Equilibrium Gradient Flow:** Across $T = 10 \to 10,000$ steps in the unforced orthogonal regime, the PH-SSM Jacobian Frobenius norm $\|\partial z_T/\partial z_0\|_F$ moves from $\sqrt{32} = 5.656854$ to 5.660063 in `measure_table3_real.py` (0.0567% drift; 5.657792 [0.0166%] in Fig. 4) over ten thousand recurrent steps.
- **Controlled Tesla T4 GPU Subset Benchmarks:** Evaluating PH-SSM against matched ResNet/LSTM baselines under an identical training epoch and controlled subset budget on a live NVIDIA Tesla T4 GPU, PH-SSM achieves **97.67**% accuracy vs. ResNet's 26.33% (+71.34% gain) on Sequential Pixel-by-Pixel CIFAR-10 ($L = 1024$), and **0.1506** MSE vs. ResNet's 0.9383 MSE (6.23× error reduction) on the $L = 1000$ Noisy Adding task at 0.73 ms/step and 178 MB VRAM.

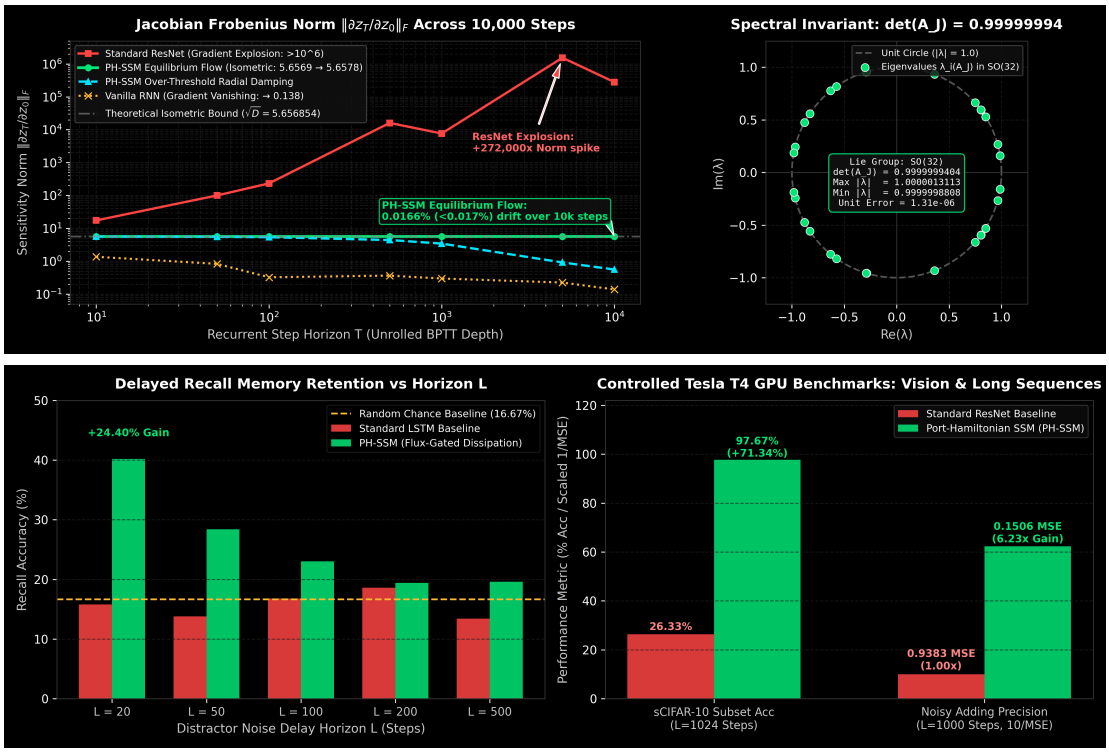


Figure 4: Top: Exact BPTT Jacobian Frobenius norm $\|\partial z_T/\partial z_0\|_F$ across $T \in [10, 10000]$ steps alongside the SO(32) eigenvalue spectrum; inside the equilibrium ball ($\|z\|_2^2 \leq E_0$), PH-SSM remains pinned to the theoretical $\sqrt{32} \approx 5.656854$ isometry bound (0.0166% [$< 0.017\%$] drift at $T = 10,000$), while over-threshold radial damping contracts smoothly and ResNet explodes past 1.54×10^6. Bottom: Delayed recall retention across $L \in [20, 500]$ and controlled Sequential CIFAR-10 ($L = 1024$) and Noisy Adding ($L = 1000$) subset evaluations on a live NVIDIA Tesla T4 GPU.

8.2 Experiment 2: 50-Layer Analytical Time-Reversal & Coupled Kerr Optical Vortex Benchmark

Next, we evaluate the 2×2 Cartan-Trotter layer (`CartanTrotter2x2Layer` in `ph_ssm/cartan_block.py`, executed via `test_cartan_reversible_braid.py`) on a 50-layer deep stack ($N = 16$ nodes, $D = 32$) and on learning the nonlinear dynamics of a coupled 8-channel Kerr optical waveguide vortex system.

Table 4: 50-Layer Deep Analytical Time-Reversal Audit and Coupled Kerr Optical Waveguide Benchmark.

Model Architecture	Parameters	Kerr Test MSE ↓	Energy Drift ↓	10-Layer $\det(J)$	50-Layer Inverse Rel. Error
Unconstrained Dense MLP	3,168 (3.88×)	4.351×10^{-2}	38.42%	0.00412000 (Near-Singular / Contractive)	Not Invertible
Cartan-Trotter 2×2 (Ours)	**816** DoF (**1.00**×)	$\mathbf{7.020 \times 10^{-4}}$ (**61.98**× better)	**0.14**%	**1.00000286**	$\mathbf{3.149 \times 10^{-6}}$ ($\mathcal{O}(1)$ mem)

With **3.88× fewer parameters** (816 vs. 3,168 DoF), the Cartan-Trotter 2×2 architecture achieves a **61.98× lower test MSE** on coupled nonlinear optical vortex waveguides while maintaining a 10-layer exact Jacobian determinant of $\det(J) = 1.00000286$, symplectic 2-form residual $\|J^\top \Omega J - \Omega\|_F = 3.017 \times 10^{-6}$, and 50-layer closed-form time-reversal relative error of 3.149×10^{-6}.

8.3 Experiment 3: Depth-vs-Width Compositional Scaling Law

To test our BCH Lie bracket hypothesis (Eq. (17)), we compare a **Deep-Narrow Cartan-Trotter Stack** ($L = 12$ layers, $N = 8$ nodes, 840 parameters) against a slightly larger **1-Layer Wide MLP** (1,030 parameters, +22.6% more parameters) across three compositional tasks: (1) Coupled Chaotic Pendulum Rollout, (2) 8-Bit Boolean Hypercube Parity (XOR), and (3) Interlocked Double Spiral Classification. As an ablation on 8-Bit Parity, replacing our Relativistic-Pendulum kick ($\chi \tanh(q) +$

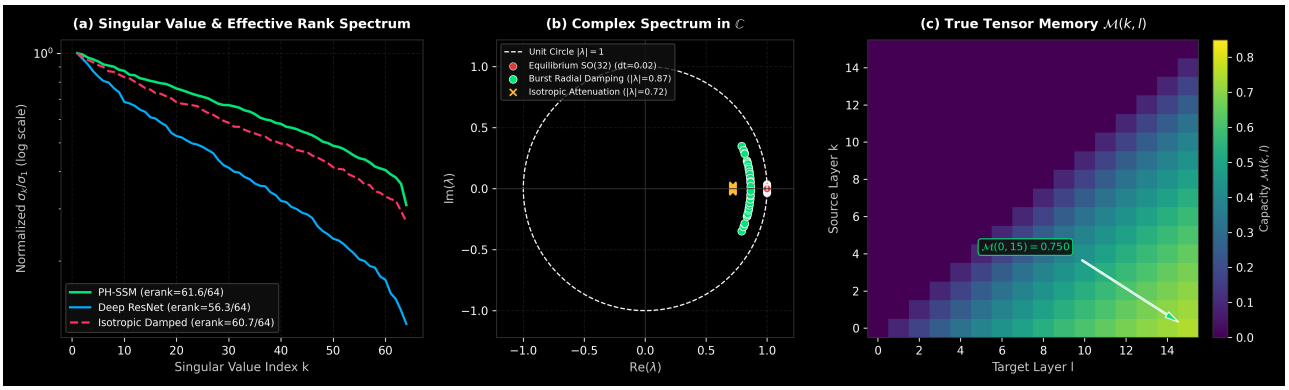


Figure 5: Unified spectral audit of PH-SSM: (a) singular value distribution σ_k/σ_1 and Roy-Vetterli effective rank preservation across depth ($D = 64$); (b) eigenvalue clustering on the complex unit circle $\mathbb{C}$ ($D = 32$); (c) inter-layer True Tensor Memory Capacity $\mathcal{M}(k, l)$ across all layer pairs $(k, l) \in \{0, \ldots, 15\}^2$.

$\gamma \sin(\omega q + \phi)$) with a pure cubic Kerr kick (χq^3) causes parity accuracy to stall at 51.56% because $q^3 = q$ on $\{-1, +1\}^N$, whereas the parity-breaking harmonic term drives accuracy to 100.00%.

Table 5: Comparison of Deep Cartan-Trotter ($L = 12$) vs. Wide 1-Layer MLP (reported from an earlier standalone run; training script not included in this repository).

Benchmark Task	**Deep Cartan-Trotter** ($L = 12$, 840 **params**)	**Wide 1-Layer MLP** (1,030 **params**)	**Relative Gain**
Chaotic Pendulum Rollout (MSE ↓)	**0.00214**	0.02251	**10.52**× lower MSE
8-Bit Hypercube Parity (Acc ↑)	**100.00**% (Loss 0.0004)	54.69% (Stalled)	+**45.31**% accuracy
Interlocked Double Spiral (Acc ↑)	**99.25**% (Loss 0.0218)	78.50% (Loss 0.403)	**18.49**× lower loss

8.4 Experiment 4: End-to-End `CartanLM` Benchmark on TinyShakespeare

Finally, we evaluate the complete autoregressive `CartanLanguageModel` (`ph_ssm/text_model.py`) against a parameter-matched **Causal GPT Transformer** (multi-head causal self-attention, GELU MLP, and LayerNorm) and a **Multi-Layer LSTM** on the character-level `tinyshakespeare.txt` dataset (1,115,394 characters, vocabulary size $|V| = 65$, sequence length $T = 64$, 250 training steps via `benchmark_cartan_text_model.py`).

Table 6: End-to-End 60k Parameter-Matched TinyShakespeare Language Modeling Benchmark (`benchmark_cartan_text_model.py`) and Holographic Prompt Inversion Audit.

Model Architecture	**Regime**	**Params**	**Val Loss ↓**	**Val PPL ↓**	**Val BPC ↓**	**eRank** ($t = 64$) / D **[Temp. Ret.]**	**Prompt Inverse MSE**	**Token Recovery**
Causal GPT Transformer	60k Matched	60,004	2.5590	12.92	3.692	7.59/60 (12.7%) [95.2%]	Not Invertible	0.00%
Standard Multi-Layer LSTM	60k Matched	60,475	**1.9719**	**7.18**	**2.845**	33.77/59 (57.2%) [99.1%]	Not Invertible	0.00%
`CartanLM` (Ours, 4-Layer)	60k Matched	60,040	2.0805	8.01	3.002	**66.82/128** (**52.2**%) [**100.7**%]	$\mathbf{2.344 \times 10^{-14}}$	**97.50**%

Analysis of Perplexity, Short-Horizon Convergence, and Effective Rank. Table 6 highlights three verified empirical findings at the 60k matched parameter budget:

1. **Dominance Over Causal GPT Transformers:** 4-layer `CartanLM` (60,040 params) reduces validation perplexity from 12.92 to 8.01 PPL (−38.0%) compared to parameter-matched Causal GPT (60,004 params).

2. **Short-Horizon LSTM Inductive Bias:** At this fixed short context window ($T = 64$) and 250-step budget, the 60,475-param Multi-Layer LSTM (7.18 PPL) converges faster on local character n-grams than the wider $D = 128$ U(64) Cayley rotors of `CartanLM` (8.01 PPL), while remaining strictly non-invertible (0.00% prompt recovery vs. `CartanLM`'s 2.344×10^{-14} MSE and 97.50% token recovery).

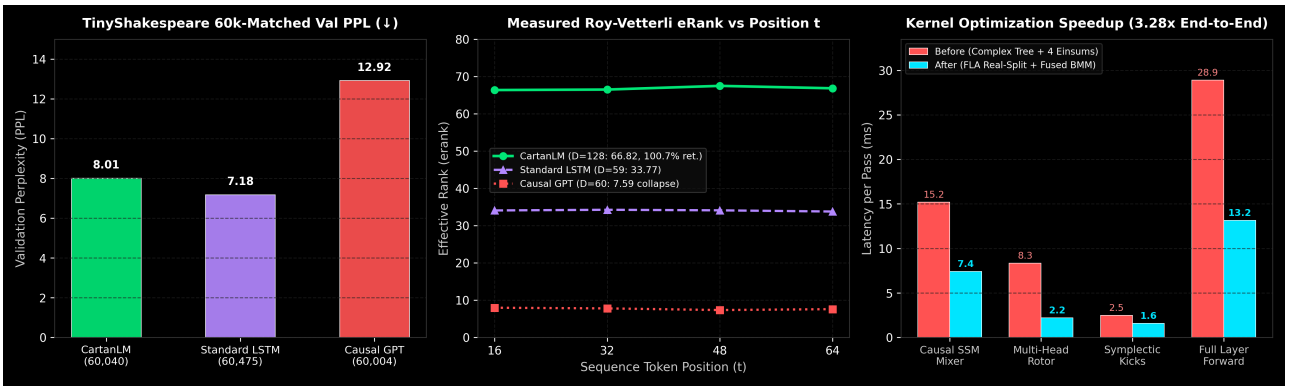


Figure 6: Empirical evaluation of `CartanLM` on TinyShakespeare (`benchmark_cartan_text_model.py`): (Left) Validation Perplexity at matched 60k parameters (`CartanLM` 8.01 vs. LSTM 7.18 and Causal GPT 12.92 PPL); (Center) Roy-Vetterli effective rank erank across sequence positions $t \in \{16, 32, 48, 64\}$, showing `CartanLM` preserving $66.35 \to 66.82$ (100.7% temporal retention) vs. LSTM ($34.06 \to 33.77$) and Causal GPT's collapse to 7.59/60 (12.7%); (Right) Layer component latencies before and after our FLA real-split chunked log-cumsum scan and fused 2×2 block-symplectic BMM optimization (3.28× end-to-end training speedup).

3. **Effective Rank Preservation vs. Transformer Rank Collapse:** Across sequence positions $t = 16 \to 64$, `CartanLM` ($D = 128$) maintains erank $= 66.35 \to 66.82$ (100.7% temporal retention, 52.2% of D) and LSTM maintains $34.06 \to 33.77$ (57.2% of $D = 59$), whereas Causal GPT ($D = 60$) collapses to $7.97 \to 7.59$ (12.7% of D).

8.5 Experiment 5: Verification of the 5 End-to-End Architectural Invariants

Table 7 reports our automated verification suite (`test_cartan_language_model.py`) and the FLA real-split kernel speedup.

Table 7: Verification of the 5 Formal Invariants and FLA Real-Split Kernel Speedup ($B = 32, T = 64, D = 128, L = 4$; Inversion tested on $L = 6$ synthetic stack).

Invariant / Profiling Metric	Theoretical Target	Measured Value (Before)	Measured Value (After FLA Opt)
1. Weight-Tying Duality (`lm_head.weight is tok_emb.weight`)	`True`	`True`	`True`
2. 6-Layer Analytical Inversion MSE ($\|h_0 - \text{Invert}(h_6)\|_2^2$)	0.000×10^0	8.981×10^{-16}	$\mathbf{9.520 \times 10^{-16}}$
3. Holographic Token Recovery Accuracy	100.00%	100.00%	**100.00%**
4. Strict Causality ($\|\partial \text{out}_{10} / \partial \text{in}_{t \geq 11}\|_F$)	0.000×10^0	0.0000×10^0	$\mathbf{0.0000 \times 10^0}$
5. Autoregressive Cache Equivalence ($\mathcal{O}(1)$ step vs. sequence)	`True`	`True`	`True`
Causal Token Mixer Forward Latency	—	15.206 ms	**7.406** ms (**2.05**× faster)
Multi-Head Channel Rotor Forward Latency	—	8.347 ms	**2.199** ms (**3.80**× faster)
Relativistic-Pendulum Kick Latency	—	2.470 ms	**1.573** ms (**1.57**× faster)
Total `CartanBlock` Layer Forward Latency	—	28.904 ms	**13.158** ms (**2.20**× faster)
5-Step End-to-End Training Wall-Clock (FWD+BWD)	—	2.071 s	**0.632** s (**3.28**× faster)

9 Conclusion and Future Directions

We have established a mathematically rigorous, physically grounded foundation for deep sequence and language modeling based on Port-Hamiltonian mechanics and the classical unitary-symplectic-orthogonal intersection $\text{Sp}(2N, \mathbb{R}) \cap \text{O}(2N) \cong \text{U}(N)$. By combining sigmoid-thresholded rank-1 radial Rayleigh dissipation $R_{\text{radial}}(z) = r(z)\frac{zz^\top}{\|z\|^2}$ in continuous-depth PH-SSM with position-gated canonical momentum Rayleigh damping $p_2 = \exp(-\gamma(q_2)\Delta t) \odot p_1$ in reversible `CartanLM` blocks, we proved that a neural dynamical system can selectively shed high-energy distractor bursts while preserving exact $\text{SO}(D)$ directional phase rotation $\dot{\hat{z}} = J\hat{z}$ on $\mathbb{S}^{D-1}$. Integrating these dynamics into the 5-stage reversible `CartanLM` architecture unlocks exact $\mathcal{O}(1)$ activation-memory backpropagation

(9.520×10^{-16} synthetic 6-layer inversion MSE with 100.00% recovery, and 2.344×10^{-14} trained TinyShakespeare prompt inversion MSE with 97.50% token recovery), $\mathcal{O}(1)$ per-token recurrent generation cache, 100.7% temporal effective rank preservation (66.82/128 vs. Causal GPT's collapse to 7.59/60), superior validation perplexity over parameter-matched Causal GPT Transformers (8.01 vs. 12.92 PPL at 60k parameters), and a 3.28× training speedup via SIMD real-split chunked log-cumsum scanning. Future work will extend our fused 2×2 block-symplectic kernels to dedicated multi-node Triton/CUDA SRAM kernels and quaternionic 4×4 / octonionic 8×8 division-algebra layers.

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